Hans Gao

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EDUCATION

San Jose State University (SJSU), San Jose, CA *Expecting Spring 2028*
Bachelor of Science in Computer Science (Transfer) | Relevant Coursework: Web Server Programming

De Anza College, Cupertino, CA *September 2024 – June 2026*
Associate in Computer Science | Relevant Coursework: Algorithms & Data Structures, Advanced C++ Programming

PROJECTS

Full-Stack Web Application (Custom Auth, Routing & Sessions) *October 2026 – Present*
Tech Stack: PHP, MySQL (PDO), JavaScript, HTML5/CSS3

- Designed and built a server-side rendered web application from scratch (no frameworks) with routing via front controller to dedicated handlers
- Implemented a login system using bcrypt password hashing and PDO prepared statements with route-level authorization checks
- Built cookie-based session management with cryptographically random tokens and hashed database checks.
- Wrote routing, authentication, sessions, and rendering code to learn abstract framework operations and develop highly-transferable skills
- In progress: handler schema, global error handler, HTML user data injection, JSON API. Planned: automated testing, security hardening, rate limiting, UI/UX, and public hosting

Durak Game AI Agent (https://github.com/Pit-tac/Durak) *November 2025*
Tech Stack: GDScript, Godot, Git, Claude Code

- Built a Godot game based on Durak, a Russian trick-taking card game
- Developed a custom Monte Carlo Tree Search algorithm with expectimax for the game AI

EXPERIENCE

Robotics Club — Operations Lead *February – June 2026*
Tech Stack: OnShape

- Designed the weapon system for a 1-lb combat robot in OnShape
- Coordinated logistics for a planned multi-college battle bot competition (Stanford, De Anza, Foothill, Mission), managing part orders, delivery timing, and inventory across all participating teams

De Anza Aerospace Club

President *November 2025 – June 2026*

- Grew membership from 4 to 15+ and led OnShape training sessions to prepare the team for the Spring 2026 two-stage rocket build
- Initiated a partnership with De Anza's manufacturing department, launching joint laser cutting workshops

Tech Lead *September 2025 – June 2026*
Tech Stack: C++ (Arduino), Onshape, OpenRocket

- Directed CAD design of the two-stage rocket, nosecone, and avionics bay in OnShape; used OpenRocket to model apogee and tune fin design for stability
- Led the electronics team in designing and programming a custom flight computer
- Proposed and led early-stage research on a Kalman filter with a 5-person team to optimize second-stage ignition timing and parachute deployment; evaluated existing EKF libraries before deciding to build a custom implementation

LEADERSHIP & ACTIVITIES

STEM Zone 2026 — De Anza Representative *April 2026*

- Presented and tabled for De Anza's Aerospace Club, Robotics Club, and Special Engineering Projects course to prospective students
- Demonstrated a robotic arm built by the Special Projects team; modified its Python control script on the spot to run a cyclical motion pattern, reducing manual operation

SKILLS

Languages: C++, PHP, JavaScript, SQL (MySQL), Python, HTML/CSS, GDScript

Tools: Git/GitHub, VS Code, Godot, Onshape, OpenRocket, Arduino, Claude Code, Cursor, GitHub Copilot

Concepts: Data Structures & Algorithms, Monte Carlo Tree Search, Expectimax, Server-Side Rendering, Authentication & Session Management, Relational Databases, Kalman Filtering, CAD & Simulation